From AF scores to clinical decisions

Oxford CDT · Machine Learning for Time Series · Clinical-utility practical

Date and time	Friday 6 November 2026, 14:30–16:00
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Demonstrator(s)	Xiao Gu, Edward Phillips
Course page	github.com/BradSegal/OxfordCDT2026ClinicalUtility

What you will do

A model gives each ECG recording a score. A service has to decide whether to send each recording for AF review. You will price the four outcomes of that decision, derive a threshold from the prices, write the function that prices a policy, and use it to choose a threshold and a model. Recording labels come from PhysioNet/CinC 2017. Every course model score is simulated. The prices are illustrative loss units.

Getting started

1. Open the course page and click the **Open in Colab** badge for the core practical.
2. Sign in with a Google account. A personal account is fine.
3. Click **File > Save a copy in Drive** and work in the copy that opens.
4. Run the setup cell. If Colab says the notebook "was not authored by Google", click **Run anyway**. Wait for `Course helpers ready.`
5. Work down the notebook in order with **Shift+Enter**. A CPU runtime is enough.

The notebooks

Notebook	When	Time
Core: From AF scores to clinical decisions	In the session	The session
Extension 1: Calibration and Platt scaling	Self-paced	About 40 min
Extension 2: Comparing models properly	Self-paced	About 45 min
Extension 3: Class weighting versus moving the threshold	Self-paced, optional	About 30 min
Extension 4: Capacity, prevalence and daily queues	Self-paced	About 35 min
Extension 5: Window-level calls and unreadable recordings	Self-paced, optional	About 40 min
Extension 6: Your own classifier's scores	Self-paced	About 20 min

Roughly: set up by 14:40, work through the core in pairs, and join the closing discussion from about 15:50. The times are rough guides; do not worry if you do not finish, as the notebook is yours to keep. Afterwards, take Extension 1 if the decision curves were hard, Extension 2 to compare models, Extension 4 for deployment, or Extension 6 with your own classifier's scores. Worked copies are in the `solutions/` folders.

How the problems work

Each problem is labelled **Predict**, **Make**, **Modify** or **Fix the bug**. Replace `...` with your answer and run the cell. The checker prints ✓ when it is right, **Not yet** with a hint that names the likely mistake, or **Not attempted yet** while the `...` remains. Problems 7 and 13 are hypotheses: the checker records your answer unmarked and the next cell shows the result. Nothing is collected or graded. If a slider does not appear, use the ordinary function call in the next cell and change its numbers. If you get a `NameError`, use **Runtime > Run before**.

Stuck for about 10 minutes? Move on. In the core, delete the # in the commented `check.use_reference(...)` line below Problems 4, 6 and 12 to use the reference version; in the extensions the same switch sits below each function a later cell uses. A cell that prints **Waiting for** names the problem it needs: once that is done or switched, click in the waiting cell, choose **Runtime > Run before**, then run the waiting cell. Worked copies are in the `solutions/` folders.

The decision matrix

	Reference AF	No reference AF
Review	True positive (TP): $C_{TP} = 1$	False positive (FP): $C_{FP} = 1$
No review	False negative (FN): $C_{FN} = 5$	True negative (TN): $C_{TN} = 0$

TP, FP, FN and TN count recordings; C_{TP}, C_{FP}, C_{FN} and C_{TN} are the price of one recording in each cell. A correct review still costs a review, so C_{TP} is not zero. C_{FP} is the whole cost of a review that finds no AF, not a surcharge on top of it.

Threshold probability

For a recording with AF probability p, review and no review have equal expected loss at

$$p_t = (C_{FP} - C_{TN}) / [\,(C_{FP} - C_{TN}) + (C_{FN} - C_{TP})\,]$$

The numerator is the harm of reviewing a recording without AF; the second bracket is the benefit of reviewing one with AF. With the default prices $p_t = 0.2$. A dearer miss lowers p_t. A score is a probability only if it is calibrated.

Net benefit

Measured against reviewing nobody, in true-positive equivalents per recording, at threshold probability p_t:

$$NB(p_t) = TP / n - (FP / n) \times p_t / (1 - p_t)$$

At $p_t = 0.2$ the weight on false positives is 0.25, the same as the priced matrix. Read a decision curve vertically: fix the preference first, then compare strategies at that value.

Categorical NRI

Reclassification from an old model to a new one at a cut, counted within each outcome group:

$$NRI_{event} = P(up \mid AF) - P(down \mid AF)$$
$$NRI_{non\text{-}event} = P(down \mid no\ AF) - P(up \mid no\ AF)$$
$$NRI = NRI_{event} + NRI_{non\text{-}event}$$

NRI adds two proportions as if they were equally valuable. It uses no prices, so a positive NRI can sit beside a worse net benefit.

Keep the denominators visible

- **Sensitivity** TP / (TP + FN): of the AF recordings, how many were reviewed?
- **Specificity** TN / (TN + FP): of those without AF, how many avoided review?
- **PPV** TP / (TP + FP): of the reviewed recordings, how many had AF?
- **NPV** TN / (TN + FN): of the unreviewed recordings, how many had no AF?

An empty denominator makes a rate undefined, not zero. "No reference AF" includes other rhythms; it does not mean healthy.

Finish with a decision

Freeze your model, cutoff and prices, and write down the evaluation loss you predict, before you look at the evaluation split. In the closing discussion, state your decision in four or five sentences: the action, your prices, your model and cutoff, the reviews and missed AF from your report, and one assumption that would change your choice.

This is teaching code. Do not use it to make decisions about real patients.